

Solution Architecture

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	5 Marks

Architecture Overview

Presentation Layer → Browser rendering Home / Predict / History pages (HTML + CSS + Chart.js), communicating via HTTP + a small JSON API.

Application Layer → Flask app (app.py) routes requests to two internal modules: **model_utils.py** (loads model.pkl, runs predictions, classifies tiers) and **db_utils.py** (all SQLite CRUD for prediction history).

Data Layer → **model.pkl** (trained scikit-learn model), **country_data.json** (per-country reference data), and **predictions.db** (SQLite history log).

Offline pipeline → train_model.py reads the UNDP CSV, trains and compares Linear Regression vs Random Forest, and exports the model + metadata + EDA graphs. seed_history.py pre-populates demo prediction history.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders 3 dashboards; sends fetch() calls to the JSON API and renders charts client-side.	HTML, CSS, Chart.js, vanilla JS
Flask Routes (app.py)	Handles page rendering and exposes /api/predict, /api/history, /api/history/, /api/history/clear.	Flask
Model Service (model_utils.py)	Loads the cached model once; runs predict_hdi() and classify_tier().	scikit-learn, joblib
Persistence Service (db_utils.py)	All reads/writes to the predictions table; isolates SQL from the rest of the app.	SQLite
Training Pipeline (train_model.py)	One-time offline job: cleans data, engineers features, trains + compares models, exports artifacts.	pandas, NumPy, scikit-learn, Matplotlib, Seaborn